

INVESTIGATION 3.1

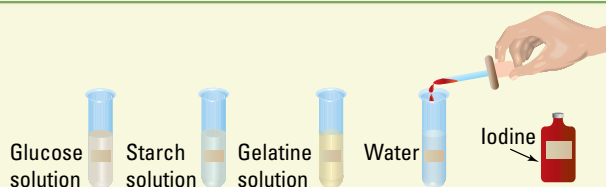
Essential testing

AIM To investigate the nutrients found in foods

Materials:

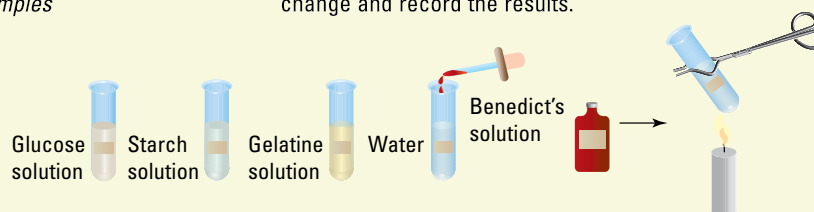
test-tube rack
4 test tubes
safety glasses
glucose solution
starch solution
gelatine solution
distilled water
iodine solution
test-tube holder
Benedict's solution
tongs
candle or Bunsen burner
matches

heatproof mat
0.01 M copper sulfate solution
1.00 M sodium hydroxide solution
food samples



GLUCOSE TEST

- Add four drops of Benedict's solution to each of the four test tubes. Gently heat each test tube over the candle or Bunsen burner flame. Observe any colour change and record the results.



METHOD AND RESULTS

- Copy and complete the table below for recording the test results.

Test results	Water	Glucose solution	Starch solution	Gelatine solution
Starch test				
Glucose test				
Protein test				

From your kitchen cupboards, select five foods to test. If they are solid, you may need to use a mortar and pestle to grind them into a 'mash' with a small amount of water before testing.

- Predict which of your food samples will contain starch, glucose and/or protein.

ESSENTIAL STANDARDS

- For each of the tests in this experiment, set up the four test tubes as shown below. After each test, clean the test tubes by rinsing with water. Make sure a fresh sample of each liquid is used for each test.

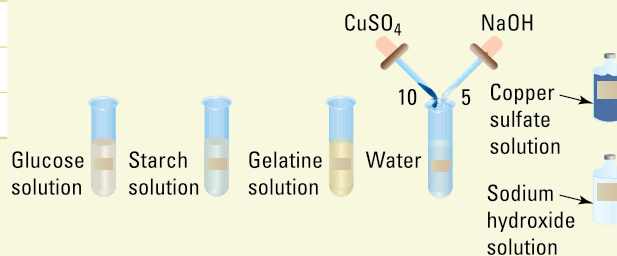


STARCH TEST

- Add two drops of iodine solution to each of the four test tubes. Observe any colour change and record the results.

PROTEIN TEST

- Add ten drops of copper sulfate solution to each of the four test tubes. Then add five drops of sodium hydroxide solution to each test tube. Observe any colour change and record the results.



ESSENTIAL FOOD TESTS

- Using the three tests above, investigate the food samples for the presence of starch, glucose and protein. (Note: Add *only* your food samples to these tests, *not* the glucose, starch or gelatine solutions.)

DISCUSS AND EXPLAIN

- Suggest why you set up standard tests and added the same volumes of solutions to each test tube.
- Which foods contain two or more of the nutrients tested for?
- Were your predictions supported by your results?
- Comment on your overall findings.
- If you were to do the food testing again, suggest how you might improve the procedure.